This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/LSP.2017.2764489, IEEE Signal Processing Letters

Learning Multi-Viewpoint Context-Aware Representation for RGB-D Scene Classification

Yingbin Zheng, Hao Ye, Li Wang, and Jian Pu

***Abstract*—Effective visual representation plays an important role in the scene classification systems. While many existing methods are focused on the generic descriptors extracted from the RGB color channels, we argue the importance of depth context, since scenes are composed with spatial variability and depth is an essential component in understanding the geometry. In this paper, we present a novel depth representation for RGB-D scene classification based on a specific designed convolutional neural network (CNN). Contrast to previous deep models that transfer from pretrained RGB CNN models, we harness model by using the multi-viewpoint depth image augmentation to overcome the data scarcity problem. The proposed CNN framework contains the dilated convolutions to expand the receptive field and a subsequent spatial pooling to aggregate multiscale contextual information. The combination of contextual design and multi-viewpoint depth images are important towards a more compact representation, compared to directly using original depth images or off-the-shelf networks. Through extensive experiments on SUN RGB-D dataset, we demonstrate that the representation outperforms recent state-of-the-arts, and combining it with standard CNN-based RGB features can lead to further improvements.**

***Index Terms*—Scene classification, RGB-D images, multi-viewpoint depth images, context-aware representation.**

I. Introduction

The construction of good image representation is a fundamental problem on a broad array of high-level vision tasks, including image recognition [1]–[4] and scene classification [5]. Great progress has been achieved with the powerful visual models from convolutional neural network (CNN) [6]. Although promising results were shown from using these representations, most of the existing methods are focused on the generic descriptors extracted from the RGB color channels. Recently, with the innovation of camera sensing systems, there exist affordable RGB-D cameras (e.g., Kinect) in the consumer market and people can take RGB images along with reliable per-pixel depth measurement. As an essential component in understanding the geometry, depth information has granted new possibilities to the computer vision community, and research problems such as action recognition [7], face recognition [8] and salient detection [9] have benefited from the robust representations learned via depth.

In this work, we tackle the problem of scene classification, which is a fundamental perception task for computer vision, robotics, and multimedia research [10]–[15]. Specifically for RGB-D scene classification, the task is to accurately classify the scene of a given image with both color and depth information. A variety of techniques have been devoted to the design of effective scene representations in the past decades. Among these methods, a few approaches exploited the spatial layout of scenes and achieved remarkable recognition results [16]–[19]. As scenes are composed with spatial variability, we believe that the depth context is also crucial for scene perception of spatial layout and understanding scenes. Recently, several works have studied the RGB-D scene classification problem [20]–[24]. In general, there are mainly two steps included in the task of these methods, i.e., learning depth features by transferring from pretrained RGB CNN models and fusion of modality or component level information. However, there still exist many limitations. A good deep model is particularly dependent on the availability of large quantities of training data. Although the recent SUN RGB-D dataset [20] is significantly larger than previous RGB-D scene datasets such as NYU depth v2 [25], the amount of RGB-D images are still order-of-magnitude smaller than RGB images. Besides, the pretrained RGB CNN models are able to successfully apply in the new RGB datasets, but are not necessarily an optimal choice for depth images.

The main point of this paper is to propose a novel depth representation for RGB-D scene classification based on a specific designed CNN with augmented depth images. Figure 1 illustrates our approach. As can be seen on the top-left side of the figure, the original RGB and depth image pairs are randomly transformed to $3D$ space and generated the multi-viewpoint depth images. The architecture, called Multi-Viewpoint Context-aware Convolutional Neural Network (MVC-CNN), takes the multi-viewpoint depth images as input and builds upon a set of contextual structures. More precisely, the dilated convolutions are applied to expand the receptive field, while a subsequent spatial pooling is used to aggregate spatial layout information. During feature extraction, the given depth image is resized into three scales, and then the concatenation of the activation of last fully connected layer from each scale is considered as the depth representation.

The advantages of our approach are in two folds. First, contrast to previous deep models that take rotated, translated, mirrored or scaled images for augmentation, we train the MVC-CNN from scratch by using the multi-viewpoint depth images to overcome the data scarcity problem. Thanks to the reconstruction ability of depth, the generated images are with different spatial variations and more suitable under RGB-D scenario. Second, the combination of contextual design and

This work was supported by the National Natural Science Foundation of China under Grants 61602459, and by the Science and Technology Commission of Shanghai Municipality under Grant 17511101902. Corresponding author: Hao Ye (e-mail: yeh@sari.ac.cn).

Y. Zheng, H. Ye, and L. Wang are with Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China.

J. Pu is with School of Computer Science and Software Engineering, East China Normal University, Shanghai, China.

1070-9908 (c) 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/LSP.2017.2764489, IEEE Signal Processing Letters

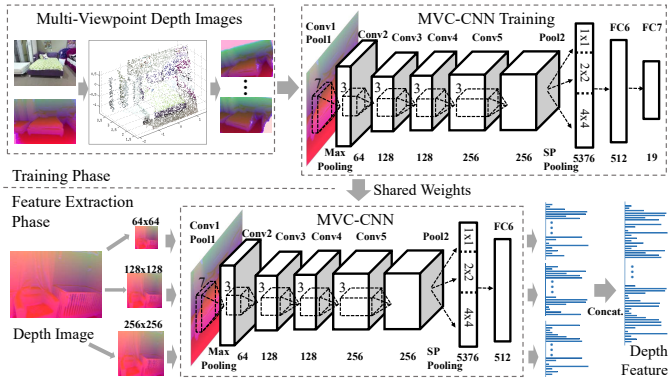

Fig. 1. Overview of the multi-viewpoint context-aware representation.

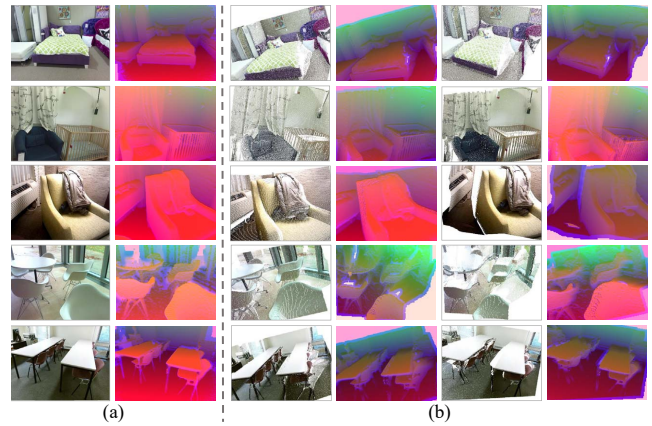

Fig. 2. Examples of multi-viewpoint depth images: (a) original color and HHA depth image; (b) transformed color and HHA depth images.

multi-viewpoint depth images are important towards a more compact and discriminant representation, compared to directly using original depth images or off-the-shelf networks. The experimental results demonstrate that our representation yields state-of-the-art performance for RGB-D scene classification, and combining it with standard CNN-based RGB features can lead to further improvements.

II. Method

A. Multi-Viewpoint Depth Images

We start by introducing the generation of multi-viewpoint depth images. The RGB-D image consists of RGB color channels and per-pixel depth measurement. Different from the color data which contains appearance information, depth indicates the scene geometry of the image. Traditional approaches to extend depth data for scene classification focus on $2D$ plane transformation such as rotation and mirror operations, and to our knowledge seldom works consider the $3D$ viewpoint transformation. Inspired by previous work of surface reconstruction [26], we aim to transform the depth data into multiple viewpoints by leveraging both color and depth, and build a larger depth image database to train better scene depth model.

Let 2-dimensional matrix $\mathbf{D}$ be the depth of a given RGB-D image, the corresponding point cloud $\mathbf{p}(\mathbf{l})$ of pixel location $\mathbf{l} = [x, y]^T$ can be mapped by

$$\mathbf{p}(\mathbf{l}) = \phi_1(\mathbf{l}, \mathbf{D}) = \begin{bmatrix} (x - c_x) \cdot \mathbf{D}(x, y)/f_x \\ (y - c_y) \cdot \mathbf{D}(x, y)/f_y \\ \mathbf{D}(x, y) \end{bmatrix}, \quad (1)$$

where c_x, c_y, f_x, f_y are the intrinsics of depth camera. The congruent transformations composed of rotation and translation are then applied to simultaneously keep the global geometry and change the viewpoint. For $\mathbf{p}(\mathbf{l})$ from the original point cloud, the transformed point $\mathbf{p}_t(\mathbf{l})$ is computed as

$$\mathbf{p}_t(\mathbf{l}) = \phi_2(\mathbf{p}(\mathbf{l}), \Theta, \mathbf{t}) = \mathbf{R}(\Theta) \cdot \mathbf{p}(\mathbf{l}) + \mathbf{t} \quad (2)$$

Here $\mathbf{t} = [t_x, t_y, t_z]^T$ is the translation vector, and the rotation matrix $\mathbf{R}(\Theta) = \mathbf{R}([\theta_x, \theta_y, \theta_z]^T) = \mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y)\mathbf{R}_z(\theta_z)$[1];

[1]The matrices are defined as $\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$, $\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$, $\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

t_* and θ_* are the transformation parameters. Since our focus is the $2D$ depth image with various viewpoint, it is necessary to have the transformed pixel location $\mathbf{l}_t$ corresponding to $\mathbf{l}$. Suppose $\mathbf{p}_t(\mathbf{l}) = [p_x, p_y, p_z]^T$, $\mathbf{l}_t$ can be calculated as the reverse of Equ. (1):

$$\mathbf{l}_t = \phi_1^{-1}(\mathbf{p}_t(\mathbf{l})) = \begin{bmatrix} p_x \cdot f_x/p_z + c_x \\ p_y \cdot f_y/p_z + c_y \end{bmatrix} \quad (3)$$

Using Equ. (1)-(3), we have

$$\mathbf{l}_t = \phi_1^{-1}(\phi_2(\phi_1(\mathbf{l}, \mathbf{D}), \Theta, \mathbf{t})) = \phi(\mathbf{l}, \mathbf{D}, \Theta, \mathbf{t}), \quad (4)$$

which means given $(\mathbf{D}, \Theta, \mathbf{t})$, each $\mathbf{l}$ can be transformed into the location $\mathbf{l}_t$. The transformation can be considered as a non-linear pixel-level warping, which is designed based the characteristic of depth data and fundamentally different from previous augmentation approaches. Comparing with learning based warping (e.g., Spatial Transform Networks [27]), our approach warps in a local manner and is more effective. Viewpoint changes are controlled by Θ and $\mathbf{t}$, which are randomly sampled from the uniform distribution $\mathcal{U}(-a, a)$; we empirically set $a = \frac{\pi}{6}$ for θ_* and $a = 1$ for t_*.

It's possible more than one original location map to a same transformed location $\mathbf{l}_t$, and we adopt the one with smallest p_z value, as the far location will be obstructed. We suppose $\mathbf{l}_t$ should fall in the same interval as $\mathbf{l}$, and therefore the pixels outside the interval will be removed. After the transformed depth image is generated, there usually exist a few unmeasured regions, and the augmentation dataset only accepts the image with measured pixels rate higher than a threshold (70%). For these remaining images, we employ the cross-bilateral filter at various scales with both transformed RGB and depth to have the inpainted depth. We apply the HHA encoding on the transformed depth, which maps the depth values with three properties of geocentric pose, i.e., horizontal disparity, height above ground, and angle of the pixel's local surface normal [28]. Examples of the multi-viewpoint depth images are displayed in Fig. 2.

B. Network Design and Training

In this section, we will introduce the design and training of proposed MVC-CNN for scene representation. The architecture of our network is summarized in Table I. In general,

1070-9908 (c) 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/LSP.2017.2764489, IEEE Signal Processing Letters

TABLE I
THE ARCHITECTURE OF PROPOSED NETWORK. HERE WE SUPPOSE THE INPUT IMAGE SIZE IS 128×128.

Type	Kernel Size / Stride / Dilation	Output Size	#Params
Conv1	7×7 / 2 / 1	$64 \times 64 \times 64$	$9K$
Pool1	3×3 / 2 / 1	$32 \times 32 \times 64$	-
Conv2	3×3 / 1 / 1	$32 \times 32 \times 128$	$74K$
Conv3	3×3 / 1 / 1	$32 \times 32 \times 128$	$147K$
Conv4	3×3 / 1 / 2	$32 \times 32 \times 256$	$295K$
Conv5	3×3 / 1 / 4	$32 \times 32 \times 256$	$590K$
Pool5	32×32 / 32 / 1 16×16 / 16 / 1 8×8 / 8 / 1	$1 \times 1 \times 256$ $2 \times 2 \times 256$ $4 \times 4 \times 256$	-
FC6	-	512	$2{,}753K$
FC7	-	19	$10K$

the network contains seven layers with weights; the first five are convolutional (Conv) and the remaining two are fully connected (FC) layers. ReLU is used as the activation function for its computational simplicity and the advantage on gradient back-propagation. The output of the last FC layer is fed to a 19-way softmax which produces a distribution over the 19 class labels of SUN RGB-D dataset we used for experiments. It is worth noticing that our model has fewer filters than the AlexNet [1] that is widely used for the scene depth features. As listed in Table I, the total parameters numbers of MVC-CNN are about $4M$, which is only 7% of AlexNet ($60M$).

The structure of our network is mainly inspired by the philosophy of VGGNet [3], and most of the convolutional layers have 3×3 filters. The down-sampling is applying directly by Conv1 and Pool1 (max-pooling) that have a stride of 2 to decrease the amount of computation. Pooling layers are usually to introduce some degree of invariance to local translation and distortion, and in this work, we embed the information with the multi-viewpoint input. Thus, less pooling layers are employed. We adopt the dilated structure to the top two convolution layers. Let $\mathbf{w}_i$ be the filter weights associated with layer Conv$i$ and $\mathbf{x}$ be the input feature map, the output $\mathbf{y}$ for location $\mathbf{l}$ with dilated d is defined as:

$$\mathbf{y}(\mathbf{l}) = \sum_{\Delta\mathbf{l} \in \mathcal{R}_d} \mathbf{w}_i(\Delta\mathbf{l}) \cdot \mathbf{x}(\mathbf{l} + \Delta\mathbf{l}) \tag{5}$$

where the grid $\mathcal{R}_d = \{(-d,-d),(-d,0),\ldots,(0,d),(d,d)\}$ for the 3×3 filters. Standard convolution is a special case when $d = 1$. Here we set the dilation of 2 and 4 for Conv4 and Conv5, respectively. The benefit of our approach is much greater representational efficiency due to the expanded receptive field size based on dilation, while the whole network is in a compact size, comparing with previous works incorporating dilated or atrous convolution (e.g., in [29], [30]).

We employ the spatial pyramid pooling (SPP) structure [31] in Pool5. Intuitively, the SPP layer is able to aggregate multiscale contextual information. More importantly, as we aim to train the network directly from the multi-viewpoint images, the pyramid structure allows us to train from coarse scale to fine with the same shared filter weights. Previous depth representations usually are extracted based on the fine-tuning of an existing CNN model learned from large-scale image datasets such as ImageNet [32] or Places database [5]. We argue that the patterns within color and depth information are diverse: while the RGB channels construct the visual appearance of an image, the HHA depth images tell more about the geometry and surface. Therefore, we train our depth network from scratch instead of using a pretrained model of traditional RGB CNN. The first step is to train an initial model using the randomly cropped 32×32 regions from multi-viewpoint depth image dataset. Then we resize all of the depth images to three specific sizes, i.e., 64×64, 128×128, and 256×256, and learn the final model by fine-tuning the weights based on the initial model. The whole training pipeline ensures the adaptability of MVC-CNN with various scales. The idea of constructing depth network from scratch has also been taken in [24] by using modified CNN architecture from AlexNet. Compared to this approach, our network focuses on modeling the context of depth, which makes our representation based on the network more discriminative than the existing works.

C. Feature Extractor and Scene Classification

We now elaborate the construction and application of depth representation from MVC-CNN. Following the below part of Fig. 1, the depth data is encoded by HHA, and then resize to the scale levels of 64×64, 128×128, and 256×256. The resized HHA depth images go through MVC-CNN, and the activations from FC6 layer are considered as the deep features of the corresponding scale. Finally, the representation is generated by concatenation over these deep features and used for RGB-D scene classification.

III. EXPERIMENTS

We perform the framework on scene categorization benchmark of SUN RGB-D dataset [20]. SUN RGB-D contains 10,335 images[2] captured by the RGB-D sensors including Microsoft Kinect and Intel RealSense. There are 19 scene categories with more than 80 images, and we follow the standard train, validate and test split in the scene categorization benchmark. The performance is evaluated by calculating the average accuracy on each category. For fair comparisons with previous RGB-D scene classification approach such as [20] and [24], we use the highly efficient LIBSVM library [35] with linear kernel throughout the experiments. We trained the MVC-CNN using SGD with a batch size of 128. The learning rate starts from 0.01 and is divided by 10 every 30 epochs. The network is implemented using pyTorch and training it on a single Nvidia GTX1080 GPU takes about 6 hours on multi-viewpoint images from SUN RGB-D.

A. Ablation Study

Extensive ablation studies are conducted to validate the efficiency of proposed depth representation.

Network. We first evaluate the effectiveness of the proposed network. Both MVC-CNN and AlexNet are tested with the same datasets and we observe significant improvements in Table III when using the MVC-CNN based representation. In addition, we also conduct the experiments without the

[2]Some images are from NYU depth v2 [25], B3DO [33], and SUN3D [34].

1070-9908 (c) 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/LSP.2017.2764489, IEEE Signal Processing Letters

TABLE II
PER-CATEGORY ACCURACY (%) OF PROPOSED MVC-CNN WITH ORIGINAL DEPTH IMAGES AND MULTI-VIEWPOINT DEPTH IMAGES.

Setting	bathroom	bedroom	classroom	computer room	conference room	corridor	dining area	dining room	discussion area	furniture store
Original depth images	68.9	52.1	**44.5**	30.8	26.7	64.2	20.8	37.0	13.5	80.0
Multi-viewpoint images	**79.5**	**55.7**	43.0	**40.0**	**37.2**	**70.9**	20.8	**60.9**	**14.4**	**80.3**

Setting	home office	kitchen	lab	lecture theatre	library	living room	office	rest space	study space	**mean**
Original depth images	**27.1**	58.7	**28.4**	**34.9**	45.7	40.8	39.1	44.2	15.8	40.7
Multi-viewpoint images	21.9	**67.7**	20.2	32.6	**58.0**	**50.4**	**49.3**	**47.9**	**16.7**	**45.6**

TABLE III
RESULTS UNDER DIFFERENT NETWORK ARCHITECTURES.

Architectures	Original depth images	Multi-viewpoint images
AlexNet [1]	37.0%	40.4%
Baseline CNN	39.4%	44.5%
MVC-CNN	40.7%	45.6%

TABLE IV
CLASSIFICATION ACCURACY BASED ON DEPTH INPUTS WITH VARIOUS SCALE LEVELS.

Scale	64	128	256	128+256	64+128+256
Accuracy (%)	40.1	41.6	42.1	43.7	**45.6**

dilation in the network structure. The *Baseline* CNN replaces dilations from Conv4 and Conv5 of MVC-CNN with standard convolutions ($d = 1$). The comparison between baseline and MVC-CNN verifies that for both original depth and multi-viewpoint images using dilated convolution as a basic element is important towards a more discriminant representation.

Multi-viewpoint. Recall that the multi-viewpoint depth images are generated from the congruent transformations, based on the warping function of $\phi(\cdot)$ shown in Equ. (4). In Table II we report the detail classification accuracy for each category trained using original depth images and the proposed multi-viewpoint images. There are 14 categories with performance gain after incorporating with multi-viewpoint and the overall accuracy increases 4.9%. Another observation is when the multi-viewpoint images are used in AlexNet and Baseline CNN, the accuracy is also improved (Table III).

Input scales. Our network is able to handle depth inputs with various scales and we also evaluate the effect of scale levels for representation. As shown in Table IV, the best results is obtained by using three scale levels (64+128+256), comparing with the representation with single scale or two scale level concatenation. We conjecture that the applied single scale may not be the one at which the network responds with optimal confidence, and multiple scale levels increase the robustness. This is consistent with the observations from recent research on object classification (e.g., [2]).

B. Comparisons with State-of-the-arts

We also study the fusion performance by combining proposed depth representations with standard CNN-based RGB features. The RGB features are based on the fine-tuning of RGB images from SUN RGB-D based on a pretrained VGGNet [3] model[3], because of its proven performance and efficiency. The fusion is done by simply concatenating the deep and feature representations. In Table V we compare with the previous best results on SUN RGB-D dataset. Fusing deep representations with color descriptors leads to an accuracy of 55.1%, which is higher than state-of-the-arts, including a semantic regularized scene classifier [21], the approaches based on feature fusion [22], [23], and a very recent depth CNN representation [24]. Fig. 3 illustrates the pre-class confusion matrix of the results using fusion of RGB CNN descriptors and proposed depth representations.

[3]Download link: https://download.pytorch.org/models/vgg16-397923af.pth

TABLE V
COMPARISONS WITH STATE-OF-THE-ARTS.

ref.	Model	Accuracy (%)		
		RGB	Depth	RGB-D
[20]	Places-CNN + RBF SVM	38.1	27.7	39.0
[21]	SS-CNN-R6	36.1	-	41.3
[22]	Multi-modal feature fusion	40.4	36.5	41.5
[23]	Modality & component fusion	40.4	36.5	48.1
[24]	RGB-D-CNN+wSVM	42.7	42.4	52.4
Ours	MVC-CNN+Linear SVM	**48.8**	**45.6**	**55.1**

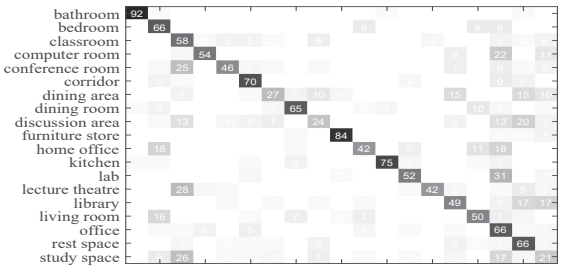

Fig. 3. Confusion matrix based on the fusion of RGB and depth features.

IV. CONCLUSION

In this paper, we have introduced a multi-viewpoint context-aware depth representation based on MVC-CNN for RGB-D scene classification. Through an extensive set of scene recognition experiments, we have shown that proposed representation is more effective than existing deep models, generating very competitive results on the SUN RGB-D benchmark. Compared with directly employing all the original HHA depth images, our approach with multi-viewpoint data offers higher performance using a much compact MVC-CNN architecture, which is designed to embed the depth context information. Moreover, we also observed that our depth model is complementary to standard CNN-based RGB features, and combining them can further boost the recognition accuracy.

1070-9908 (c) 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/LSP.2017.2764489, IEEE Signal Processing Letters

REFERENCES

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems (NIPS)*, 2012, pp. 1097–1105.

[2] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun, "Overfeat: Integrated recognition, localization and detection using convolutional networks," in *International Conference on Learning Representations (ICLR)*, 2014.

[3] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *International Conference on Learning Representations (ICLR)*, 2015.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[5] B. Zhou, A. Lapedriza, J. Xiao, A. Torralba, and A. Oliva, "Learning deep features for scene recognition using places database," in *Advances in Neural Information Processing Systems (NIPS)*, 2014, pp. 487–495.

[6] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[7] Y. Song, S. Liu, and J. Tang, "Describing trajectory of surface patch for human action recognition on rgb and depth videos," *IEEE Signal Processing Letters*, vol. 22, no. 4, pp. 426–429, 2015.

[8] T. Mantecn, C. R. del Blanco, F. Jaureguizar, and N. Garcła, "Visual face recognition using bag of dense derivative depth patterns," *IEEE Signal Processing Letters*, vol. 23, no. 6, pp. 771–775, 2016.

[9] A. Wang and M. Wang, "Rgb-d salient object detection via minimum barrier distance transform and saliency fusion," *IEEE Signal Processing Letters*, vol. 24, no. 5, pp. 663–667, 2017.

[10] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2006, pp. 2169–2178.

[11] L.-J. Li, H. Su, L. Fei-fei, and E. P. Xing, "Object bank: A high-level image representation for scene classification semantic feature sparsification," in *Advances in Neural Information Processing Systems (NIPS)*, 2010, pp. 1378–1386.

[12] M. Pandey and S. Lazebnik, "Scene recognition and weakly supervised object localization with deformable part-based models," in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 1307–1314.

[13] Y. Zheng, Y.-G. Jiang, and X. Xue, "Learning hybrid part filters for scene recognition," in *European Conference on Computer Vision (ECCV)*, 2012, pp. 172–185.

[14] M. Juneja, A. Vedaldi, C. V. Jawahar, and A. Zisserman, "Blocks that shout: Distinctive parts for scene classification," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 923–930.

[15] D. Lin, C. Lu, R. Liao, and J. Jia, "Learning important spatial pooling regions for scene classification," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 3726–3733.

[16] A. Oliva and A. Torralba, "Modeling the shape of the scene: A holistic representation of the spatial envelope," *International Journal of Computer Vision*, vol. 42, no. 3, pp. 145–175, 2001.

[17] J. Vogel and B. Schiele, "Semantic modeling of natural scenes for content-based image retrieval," *International Journal of Computer Vision*, vol. 72, no. 2, pp. 133–157, 2007.

[18] H. Lu, Y. Zheng, X. Xue, and Y. Zhang, "Content and context-based multi-label image annotation," in *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2009, pp. 61–68.

[19] C. Weng, H. Wang, J. Yuan, and X. Jiang, "Discovering class-specific spatial layouts for scene recognition," *IEEE Signal Processing Letters*, vol. 24, no. 8, pp. 1143–1147, 2017.

[20] S. Song, S. P. Lichtenberg, and J. Xiao, "Sun rgb-d: A rgb-d scene understanding benchmark suite," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 567–576.

[21] Y. Liao, S. Kodagoda, Y. Wang, L. Shi, and Y. Liu, "Understand scene categories by objects: A semantic regularized scene classifier using convolutional neural networks," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 2318–2325.

[22] H. Zhu, J.-B. Weibel, and S. Lu, "Discriminative multi-modal feature fusion for rgbd indoor scene recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2969–2976.

[23] A. Wang, J. Cai, J. Lu, and T.-J. Cham, "Modality and component aware feature fusion for rgb-d scene classification," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5995–6004.

[24] X. Song, L. Herranz, and S. Jiang, "Depth cnns for rgb-d scene recognition: Learning from scratch better than transferring from rgb-cnns," in *AAAI Conference on Artificial Intelligence (AAAI)*, 2017, pp. 4271–4277.

[25] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, "Indoor segmentation and support inference from rgbd images," in *European Conference on Computer Vision (ECCV)*, 2012, pp. 746–760.

[26] R. A. Newcombe, S. Izadi, O. Hilliges, D. Molyneaux, D. Kim, A. J. Davison, P. Kohi, J. Shotton, S. Hodges, and A. Fitzgibbon, "Kinectfusion: Real-time dense surface mapping and tracking," in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2011, pp. 127–136.

[27] M. Jaderberg, K. Simonyan, A. Zisserman, and K. Kavukcuoglu, "Spatial transformer networks," in *Advances in Neural Information Processing Systems (NIPS)*, 2015, pp. 2017–2025.

[28] S. Gupta, R. Girshick, P. Arbeláez, and J. Malik, "Learning rich features from rgb-d images for object detection and segmentation," in *European Conference on Computer Vision (ECCV)*, 2014, pp. 345–360.

[29] F. Yu, V. Koltun, and T. Funkhouser, "Dilated residual networks," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[30] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2017.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 37, no. 9, pp. 1904–1916, 2015.

[32] J. Deng, W. Dong, R. Socher, L. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.

[33] A. Janoch, S. Karayev, Y. Jia, J. T. Barron, M. Fritz, K. Saenko, and T. Darrell, "A category-level 3d object dataset: Putting the kinect to work," in *Consumer Depth Cameras for Computer Vision*, 2013, pp. 141–165.

[34] J. Xiao, A. Owens, and A. Torralba, "Sun3d: A database of big spaces reconstructed using sfm and object labels," in *IEEE International Conference on Computer Vision (ICCV)*, 2013, pp. 1625–1632.

[35] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Transactions on Intelligent Systems and Technology*, vol. 2, pp. 27:1–27:27, 2011.

1070-9908 (c) 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.